

Unit B (Generalized Linear Models)

Mean and variance I

Let us recall that $p(y_i; \theta_i, \phi) = \exp\left\{ \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\}$, so that

$$e(\theta_i, \phi; y_i) = \log p(y_i; \theta_i, \phi) = \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi)$$

which is the log-likelihood for the i -th contribution, or the loglik when $n=1$.

Then, we can compute the **score function**

$$\begin{aligned} \frac{\partial}{\partial \theta_i} e(\theta_i, \phi; y_i) &= \frac{\partial}{\partial \theta_i} \left[\frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + \cancel{c(y_i, \phi)} \right] \\ &= \frac{y_i - b'(\theta_i)}{a_i(\phi)}, \quad \text{with } b'(\theta_i) = \frac{\partial}{\partial \theta_i} b(\theta) \\ &\quad \text{first derivative of } b(\theta) \end{aligned}$$

The **second derivative** is equal to

$$\begin{aligned} \frac{\partial^2}{\partial \theta_i^2} e(\theta_i, \phi; y_i) &= \frac{\partial}{\partial \theta_i} \left[\frac{\partial}{\partial \theta_i} e(\theta_i, \phi; y_i) \right] = \frac{\partial}{\partial \theta_i} \frac{y_i - b'(\theta_i)}{a_i(\phi)} = \\ &= - \frac{b''(\theta_i)}{a_i(\phi)}, \quad \text{with } b''(\theta_i) = \frac{\partial}{\partial \theta_i} b'(\theta_i) = \frac{\partial^2}{\partial \theta_i^2} b(\theta_i) \\ &\quad \text{second derivative of } b(\theta_i). \end{aligned}$$

Recall the fundamental identities (valid for regular models, such as ED).

$$\mathbb{E} \left[\frac{\partial}{\partial \theta_i} e(\theta_i, \phi; Y_i) \right] = 0 \quad (\text{The score is equal to zero})$$

$$\Rightarrow \mathbb{E} \left[\frac{y_i - b'(\theta_i)}{a_i(\phi)} \right] = 0 \Rightarrow \frac{\mathbb{E}[y_i] - b'(\theta_i)}{\cancel{a_i(\phi)}} = 0 \Rightarrow \mathbb{E}[y_i] = b'(\theta_i).$$

↖ Positive constant ↳ Does not depend on ϕ !

Moreover, the second fundamental identity is

$$E\left[\left(\frac{\partial}{\partial \theta_i} e(\theta_i, \phi; y_i)\right)^2\right] = E\left[-\frac{\partial^2}{\partial^2 \theta_i} e(\theta_i, \phi; y_i)\right]$$

$$\stackrel{''}{=} \text{var}\left(\frac{\partial}{\partial \theta_i} e(\theta_i, \phi; y_i)\right)$$

because the expectation is 0.

$$\Rightarrow E\left[\left(\frac{y_i - b'(\theta_i)}{a_i(\phi)}\right)^2\right] = E\left[\frac{b''(\theta_i)}{a_i(\phi)}\right]$$

$\uparrow$ There is nothing random here

$$\Rightarrow \text{var}\left(\frac{y_i - b'(\theta_i)}{a_i(\phi)}\right) = \frac{b''(\theta_i)}{a_i(\phi)} \Rightarrow \frac{\text{var}(y_i)}{a_i(\phi)^2} = \frac{b''(\theta_i)}{a_i(\phi)}$$

$$\Rightarrow \text{var}(y_i) = \frac{a_i(\phi)^2}{a_i(\phi)} b''(\theta_i) = a_i(\phi) b''(\theta_i).$$

Note that

$$b''(\theta_i) = \frac{\text{var}(y_i)}{a_i(\phi)} > 0 \Rightarrow b(\cdot) \text{ is } \underline{\text{convex}}, b'(\cdot) \text{ is}$$

$\downarrow$
Second derivative, $b(\cdot)$



monotone increasing.

Gaussian distribution

Let $y_i \sim N(\mu_i, \sigma^2)$. The density function of y_i is

$$\begin{aligned} p(y_i; \mu_i, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} (y_i - \mu_i)^2\right\}. \\ &\quad \text{expand it!} \\ &= \exp\left\{-\frac{1}{2\sigma^2} (y_i^2 - 2y_i\mu_i + \mu_i^2) - \log(\sqrt{2\pi\sigma^2})\right\} \\ &= \exp\left\{-\frac{1}{2\sigma^2} y_i^2 + \frac{y_i\mu_i}{\sigma^2} - \frac{\mu_i^2/2}{\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)\right\} \\ &= \exp\left\{\underbrace{\frac{1}{\sigma^2} y_i \mu_i}_{\phi \nearrow \theta_i} - \underbrace{\frac{\mu_i^2/2}{\sigma^2}}_{b(\theta_i)} - \underbrace{\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} y_i^2}_{c(y_i, \phi)}\right\} \\ &= \exp\left\{\frac{1}{\phi} (y_i \theta_i - \theta_i^2/2) + c(y_i, \phi)\right\}. \end{aligned}$$

Therefore, we have

$\theta_i = \mu_i \Rightarrow$ *mean and natural parametrization coincide*

$$a_i(\phi) = \phi = \sigma^2$$

$$b_i(\theta_i) = \frac{\theta_i^2}{2}$$

Moreover, using the results developed before, we obtain:

$$\frac{\partial}{\partial \theta_i} b(\theta_i) = \frac{\partial}{\partial \theta_i} \frac{\theta_i^2}{2} = \frac{2}{2} \theta_i = \theta_i = \mathbb{E}[y_i]$$

$$\frac{\partial^2}{\partial \theta_i^2} b(\theta_i) = \frac{\partial}{\partial \theta_i} b'(\theta_i) = \frac{\partial}{\partial \theta_i} \theta_i = 1. \Rightarrow \text{var}(\mu_i) = 1 \text{ and}$$

$$\text{var}(y_i) = a_i(\phi) b''(\theta_i) = \phi \cdot 1 = \sigma^2.$$

Poisson distribution

Let $y_i \sim \text{Poisson}(\mu_i)$. The density function is

$$p(y_i; \mu_i) = \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!}$$

$$\theta_i = \log(\mu_i)$$

$$\mu_i = e^{\theta_i}$$

$$= \exp\{y_i \underbrace{\log(\mu_i)}_{=\theta_i} - \mu_i - \log(y_i!)\}$$

$$= \exp\{y_i \theta_i - e^{\theta_i} - c(y_i)\}$$

From here, we get

$$a_i(\phi) = 1, \quad b(\theta_i) = e^{\theta_i}, \quad c(y_i) = -\log(y_i!)$$

$$\Rightarrow b'(\theta_i) = b''(\theta_i) = e^{\theta_i}$$

$$\Rightarrow E[y_i] = b'(\theta_i) = e^{\theta_i},$$

$$\text{var}(y_i) = \underbrace{a_i(\phi)}_1 b''(\theta_i) = e^{\theta_i} = \mu_i,$$

meaning that $v(\mu_i) = \mu_i$.

Gamma distribution

Let $y_i \sim \text{Gamma}(\alpha, \lambda_i)$. The density function is

$$p(y_i; \alpha, \lambda_i) = \frac{1}{\Gamma(\alpha)} \lambda_i^\alpha y_i^{\alpha-1} e^{-\lambda_i y_i}, \quad y > 0$$

$$= \exp\{\alpha \log(\lambda_i) + (\alpha-1) \log(y_i) - \lambda_i y_i - \log \Gamma(\alpha)\}$$

$$= \exp\left\{ \underbrace{\alpha}_{\substack{\text{"1/\phi"} \\ \phi}} \left[\log(\lambda_i) - \underbrace{\frac{\lambda_i}{\alpha}}_{\theta_i} y_i \right] + (\alpha-1) \log(y_i) - \log \Gamma(\alpha) \right\}$$

$$\alpha = \frac{1}{\phi}, \quad \phi = \frac{1}{\alpha}$$

$$\theta_i = -\frac{\lambda_i}{\alpha}$$

$$\lambda_i = -\theta_i \alpha = -\theta_i / \phi$$

$$= \exp\left\{ \frac{1}{\phi} \left[\underbrace{\log(-\theta_i / \phi)}_{\log(-\theta_i) - \log(\phi)} + \theta_i y_i \right] + \left(\frac{1}{\phi} - 1 \right) \log(y_i) - \log \Gamma\left(\frac{1}{\phi}\right) \right\}$$

$$= \exp\left\{ \frac{1}{\phi} \left[\theta_i y_i - (-\log(-\theta_i)) \right] - \frac{1}{\phi} \log(\phi) + \left(\frac{1}{\phi} - 1 \right) \log(y_i) - \log \Gamma\left(\frac{1}{\phi}\right) \right\}.$$

$$= c(y_i, \phi).$$

$$= \exp\left\{ \frac{\theta_i y_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\},$$

with

$$b(\theta_i) = -\log(-\theta_i),$$

$$\theta_i < 0 \text{ and } a_i(\phi) = \phi$$

Thus:

$$E[y_i] = b'(\theta_i) = \frac{\partial}{\partial \theta_i} -\log(-\theta_i) = -\frac{1}{\theta_i} = \mu_i.$$

Careful with the "triple" negative sign.

$$\text{var}(y_i) = \underbrace{a_i(\phi)}_{\phi} b''(\theta_i) = \phi \frac{\partial}{\partial \theta_i} -\frac{1}{\theta_i} = \phi \frac{1}{\theta_i^2} = \phi \underbrace{\mu_i^2}_{\text{var}(\mu_i)}$$

Binomial distribution

Let $S_i \sim \text{Binomial}(m_i, \pi_i)$. The density of S_i is

$$p_s(s; m_i, \pi_i) = \binom{m_i}{s} \pi_i^s (1 - \pi_i)^{m_i - s}, \quad s = 0, 1, \dots, m_i.$$

In GLTs, we are interested in modelling $y_i = \frac{S_i}{m_i}$, whose density is

$$\begin{aligned} p_y(y_i; m_i, \pi_i) &= P[Y_i = y_i] = P\left[\frac{S_i}{m_i} = y_i\right] = P[S_i = y_i m_i] \\ &= p_s(y_i m_i; m_i, \pi_i), \quad y_i \in \left\{0, \frac{1}{m_i}, \frac{2}{m_i}, \dots, \frac{m_i}{m_i}\right\} \end{aligned}$$

$$= \binom{m_i}{m_i y_i} \pi_i^{y_i m_i} (1 - \pi_i)^{m_i - m_i y_i}$$

$$= \binom{m_i}{m_i y_i} \left(\frac{\pi_i}{1 - \pi_i}\right)^{y_i m_i} (1 - \pi_i)^{m_i}$$

$$= \exp\left\{ m_i y_i \underbrace{\log\left(\frac{\pi_i}{1 - \pi_i}\right)}_{\theta_i} + m_i \log(1 - \pi_i) + \underbrace{\log\left(\binom{m_i}{m_i y_i}\right)}_{c(y_i)} \right\}$$

$\eta_i(\phi) = \frac{1}{m_i}$

$$\theta_i = \log\left(\frac{\pi_i}{1 - \pi_i}\right) \Rightarrow e^{\theta_i} = \frac{\pi_i}{1 - \pi_i} \Rightarrow e^{\theta_i} - \pi_i e^{\theta_i} = \pi_i$$

Note that $\pi_i = E[Y_i] = \mu_i$.

$$\Rightarrow \pi_i + \pi_i e^{\theta_i} = e^{\theta_i}$$

$$\Rightarrow \pi_i (1 + e^{\theta_i}) = e^{\theta_i}$$

$$\left| \pi_i = \frac{e^{\theta_i}}{1 + e^{\theta_i}} \right| \Rightarrow 1 - \pi_i = \frac{1}{1 + e^{\theta_i}}$$

$$= \exp\left\{ m_i y_i \theta_i + m_i \log\left[\frac{1}{1 + e^{\theta_i}}\right] + c(y_i, m_i) \right\}$$

$$= \exp \left\{ \frac{y_i \theta_i - \log(1 + e^{\theta_i})}{1/m_i} + c(y_i, m_i) \right\}, \text{ therefore}$$

$$a_i(\phi) = \frac{1}{m_i} \text{ (No dispersion parameter)}; \quad b(\theta_i) = \log(1 + e^{\theta_i}).$$

From here, we establish that

$$E[y_i] = b'(\theta_i) = \frac{\partial}{\partial \theta_i} \log(1 + e^{\theta_i}) = \frac{e^{\theta_i}}{1 + e^{\theta_i}} = \tilde{\pi}_i = \mu_i$$

$$\text{var}(y_i) = \underbrace{a_i(\phi)}_{1/m_i} b''(\theta_i) = \frac{1}{m_i} \frac{\partial}{\partial \theta_i} \frac{e^{\theta_i}}{1 + e^{\theta_i}}$$

$$= \frac{1}{m_i} \left[\frac{e^{\theta_i} \cdot (1 + e^{\theta_i}) - e^{\theta_i} e^{\theta_i}}{(1 + e^{\theta_i})^2} \right] =$$

$$= \frac{1}{m_i} \frac{e^{\theta_i}}{(1 + e^{\theta_i})^2} = \frac{1}{m_i} \frac{e^{\theta_i}}{1 + e^{\theta_i}} \cdot \frac{1}{1 + e^{\theta_i}}$$

$$= \frac{1}{m_i} \tilde{\pi}_i (1 - \tilde{\pi}_i) = \frac{1}{m_i} \underbrace{\mu_i (1 - \mu_i)}_{= v(\mu_i)}$$

Likelihood quantities

Suppose $y_i \stackrel{\text{ind}}{\sim} \text{ED}(\mu_i, a_i(\phi) v(\mu_i))$ be the response variable of a GLM, with $g(\mu_i) = x_i^T \beta$. The joint distribution of $(y_1, \dots, y_m)$ is

$$p(y; \beta, \phi) = \prod_{i=1}^m p(y_i; \beta, \phi) = \prod_{i=1}^m \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\}$$

Independence

where $\theta_i = \theta(\mu_i) = \theta(g^{-1}(x_i^T \beta))$. The log-likelihood therefore is

$$e(\beta, \phi) = \log p(y; \beta, \phi) = \log \left[\prod_{i=1}^m \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\} \right]$$

properties of logarithms

$$= \sum_{i=1}^m \log \left[\exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\} \right]$$

$$= \sum_{i=1}^m \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi)$$

Likelihood equations

To find the MLE, we want to set to zero the first derivative (score function):

$$e_*(\beta; \phi) := \frac{\partial}{\partial \beta} e(\beta, \phi) = \begin{bmatrix} e_1(\beta, \phi) \\ \vdots \\ e_p(\beta, \phi) \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} e(\beta, \phi) \\ \vdots \\ \frac{\partial}{\partial \beta_p} e(\beta, \phi) \end{bmatrix}$$

For inference, we will also need to compute the matrix of second derivatives, with the sign changed

$$J = - \frac{\partial}{\partial \beta} \frac{\partial}{\partial \beta^T} e(\beta, \phi) = - \begin{bmatrix} \frac{\partial^2}{\partial \beta_1^2} e(\beta, \phi) & \dots & \frac{\partial^2}{\partial \beta_1 \partial \beta_p} e(\beta, \phi) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial \beta_p \partial \beta_1} e(\beta, \phi) & \dots & \frac{\partial^2}{\partial \beta_p^2} e(\beta, \phi) \end{bmatrix}$$

At the moment, suppose ϕ is unknown (this is why we do not take the derivatives):

Score function

$$e_2(\beta, \phi) = \frac{\partial}{\partial \beta_2} e(\beta, \phi) = \frac{\partial}{\partial \beta_2} \sum_{i=1}^m \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i; \phi), \quad r = 1, \dots, p.$$

$$= \sum_{i=1}^m \frac{\partial}{\partial \beta_2} \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + \cancel{\frac{\partial}{\partial \beta_2} c(y_i; \phi)}$$

$$= \sum_{i=1}^m \frac{1}{a_i(\phi)} \left[y_i \frac{\partial \theta_i}{\partial \beta_2} - \frac{\partial b(\theta_i)}{\partial \beta_2} \right]$$

Now, note that

$$\frac{\partial b(\theta_i)}{\partial \beta_2} = b'(\theta_i) \cdot \frac{\partial \theta_i}{\partial \beta_2} = \mu_i \cdot \frac{\partial \theta_i}{\partial \beta_2}$$

$$= \sum_{i=1}^m \frac{1}{a_i(\phi)} (y_i - \mu_i) \cdot \frac{\partial \theta_i}{\partial \beta_2}, \quad r = 1, \dots, p$$

Note Recall that $a_i(\phi) = \phi / w_i$, therefore the solution of the score is

$$\cancel{\frac{1}{\phi}} \sum_{i=1}^m w_i (y_i - \mu_i) \frac{\partial \theta_i}{\partial \beta_2} = 0, \quad r = 1, \dots, p.$$

The parameter simplifies! From here, it is a bit long. Recall that

$\theta_i = \theta(\mu_i)$, which is the inverse of $\mu(\theta_i)$

$$g(\mu_i) = \eta_i = x_i^T \beta \Rightarrow \mu_i = g^{-1}(\eta_i)$$

As an application of the chain rule of derivation for composite functions

$$\frac{\partial \theta_i}{\partial \beta_2} = \frac{\partial \theta_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta_2}$$

We now apply The lemma for inverse functions.

Lemma. Let $f(x)$ be a function with inverse $g(x) = f^{-1}(x)$ and first derivative $f'(x)$. Then

$$g'(x) = \frac{\partial g}{\partial x} = [f^{-1}]'(x) = \frac{1}{f'(f^{-1}(x))}.$$

We can use this lemma:

$$\frac{\partial \theta_i}{\partial \mu_i} = \theta'(\mu_i) = \frac{1}{\mu'(\theta(\mu_i))} = \frac{1}{b''(\theta(\mu_i))} = \frac{1}{v(\mu_i)}.$$

From the above lemma we proved this!

Moreover:

$$\frac{\partial \mu_i}{\partial \eta_i} = \frac{\partial g^{-1}(\eta_i)}{\partial \eta_i} = \frac{1}{g'(g^{-1}(\eta_i))} = \frac{1}{g'(\mu_i)}$$

Finally

$$\frac{\partial \eta_i}{\partial \beta_2} = \frac{\partial}{\partial \beta_2} x_i^T \beta = \frac{\partial}{\partial \beta_2} (x_{i1} \beta_1 + \dots + x_{i2} \beta_2 + \dots + x_{ip} \beta_p) = x_{i2}$$

Putting things together:

$$\frac{\partial \theta_i}{\partial \beta_2} = \frac{\partial \theta_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta_2} = \frac{1}{v(\mu_i)} \cdot \frac{1}{g'(\mu_i)} \cdot x_{i2}$$

And so...

$$\frac{\partial}{\partial \beta_2} e(\beta, \phi) = \sum_{i=1}^m \frac{1}{a_i(\phi)} (y_i - \mu_i) \left(\frac{\partial \theta_i}{\partial \beta_2} \right)$$
$$= \sum_{i=1}^m \frac{1}{a_i(\phi)} \frac{(y_i - \mu_i)}{v(\mu_i)} \cdot \frac{x_{i2}}{g'(\mu_i)}$$

we know its value

$$= \sum_{i=1}^m \frac{(y_i - \mu_i)}{var(y_i)} \cdot \frac{x_{i2}}{g'(\mu_i)}$$

(See slides for matrix notation).

Canonical link

In the canonical link case, we have

$$\theta(\mu_i) = g(\mu_i) \Rightarrow \theta_i = x_i^T \beta.$$

Therefore

$$\frac{\partial \theta_i}{\partial \mu_i} = \frac{1}{v(\mu_i)} = g'(\mu_i)$$

(or always)

By definition

$$v(\mu_i) g'(\mu_i) = 1.$$

$$\Rightarrow \frac{\partial \theta_i}{\partial \beta_2} = x_{i2}, \text{ which can also be computed directly, since } \theta_i = x_i^T \beta.$$

In the canonical link, the first derivative is

$$e_2(\beta, \phi) = \sum_{i=1}^m \frac{1}{a_i(\phi)} (y_i - \mu_i) x_{i2} = \frac{1}{\phi} \sum_{i=1}^m w_i (y_i - \mu_i) x_{i2}.$$

In matrix notation, we have

$$e_*(\beta, \phi) = X^T \Omega (y - \mu), \Rightarrow X^T (y - \mu) \text{ if } \Omega = I_m.$$

Observed and expected information

The second derivative can also be obtained (with some work). The generic element $[j]_{rs}$ is

$$j_{rs} = - \frac{\partial}{\partial \beta_s} \frac{\partial}{\partial \beta_r} \ell(\beta, \phi) = - \frac{\partial}{\partial \beta_s} \ell_r(\beta, \phi)$$

First derivative

$$= - \frac{\partial}{\partial \beta_s} \left[\sum_{i=1}^m \frac{1}{a_i(\phi)} (y_i - \mu_i) \frac{\partial \theta_i}{\partial \beta_r} \right]$$

$$= \sum_{i=1}^m \frac{1}{a_i(\phi)} \frac{\partial}{\partial \beta_s} \overbrace{(\mu_i - y_i)}^A \overbrace{\frac{\partial \theta_i}{\partial \beta_r}}^B$$

$$= \sum_{i=1}^m \frac{1}{a_i(\phi)} \frac{\partial \mu_i}{\partial \beta_s} \frac{\partial \theta_i}{\partial \beta_r} + (\mu_i - y_i) \frac{\partial \theta_i}{\partial \beta_s \partial \beta_r} \Bigg], \quad r, s = 1, \dots, p.$$

Derivative of a product A · B.

It is possible to get the derivative of $\frac{\partial \theta_i}{\partial \beta_s \partial \beta_r}$ but it is not necessary, because we need

$$i_{rs} = \mathbb{E}[j_{rs}] \quad (\text{Expected information})$$

$$i_{rs} = \sum_{i=1}^m \frac{1}{a_i(\phi)} \frac{\partial \mu_i}{\partial \beta_s} \frac{\partial \theta_i}{\partial \beta_r} + (\mu_i - \mathbb{E}[y_i]) \frac{\partial \theta_i}{\partial \beta_s \partial \beta_r}$$

this is constant

Moreover, from the previous calculations:

$$\frac{\partial \theta_i}{\partial \beta_r} = \frac{1}{v(\mu_i) g'(\mu_i)} x_{ir}, \quad \frac{\partial \mu_i}{\partial \beta_s} = \frac{1}{g'(\mu_i)} x_{is}$$

$$\Rightarrow i_{rs} = \sum_{i=1}^m \frac{1}{a_i(\phi)} \frac{1}{v(\mu_i) g'(\mu_i)^2} x_{ir} x_{is}, \quad r, s = 1, \dots, p.$$

We can organize this into a matrix:

$$I = X^T W X, \text{ with } W = \text{diag}(w_1, \dots, w_m).$$

$$w_i = \frac{1}{e_i(\phi) \sqrt{p_i} g'(p_i)^2} = \frac{1}{\text{var}(y_i) g'(p_i)^2}.$$

Comomic link

In the canonical link, we have $\theta_i = x_i^T \beta$ and so

$$\frac{\partial^2 \Theta_i}{\partial \beta_r \partial \beta_s} = 0 \Rightarrow i_{rs} = \gamma_{rs} \Rightarrow \text{The matrix } \gamma_{rs} \text{ is non-stochastic!}$$

→ This terms appear in the calculations of j_{23} (see above).

Moreover $v(\mu_i) = 1/g'(\mu_i)$.

$$\begin{aligned} i_{23} &= \sum_{i=1}^3 \frac{1}{a_i(\phi)} \frac{1}{v(\mu_i) g'(\mu_i)^2} x_{i2} x_{i3} = \\ &= \sum_{i=1}^3 \frac{1}{a_i(\phi)} \frac{1}{v(\mu_i)} x_{i2} x_{i3} \end{aligned}$$

$$\Rightarrow I = X^T W X, \quad w_i = \frac{v(p_i)}{a_i(p_i)}$$

IRLS algorithm

The IRLS algorithm updates the coefficients as follows:

$$\beta^{(t+1)} = \beta^{(t)} + (\mathbf{I}^{(t)})^{-1} e_*(\beta^{(t)}).$$

We can simplify these formulas a little bit. First:

$$\begin{aligned} e_2(\beta) &= \sum_{i=1}^m \frac{1}{a_i(\beta)} \frac{(y_i - \mu_i)}{v(\mu_i)} \frac{x_{i2}}{g'(\mu_i)} \frac{g'(\mu_i)}{g'(\mu_i)} \\ &= \sum_{i=1}^m w_i x_{i2} \underbrace{(y_i - \mu_i) g'(\mu_i)}_{u_i} \end{aligned}$$

w_i
 u_i

In matrix notation, we can write

$$\begin{aligned} e_*(\beta^{(t)}) &= \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{u}^{(t)}, & \mathbf{u}^{(t)} &= (u_1^{(t)}, \dots, u_m^{(t)}), \\ \mathbf{I}^{(t)} &= \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{u}^{(t)}. & u_i^{(t)} &= (y_i - \mu_i^{(t)}) g'(\mu_i^{(t)}) \end{aligned}$$

Combining the above equations, we get

$$\beta^{(t+1)} = \beta^{(t)} + (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{u}^{(t)}.$$

$$\begin{aligned} (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X}) \beta^{(t+1)} &= (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X}) \beta^{(t)} + \cancel{(\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})} \cancel{(\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1}} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{u}^{(t)} \\ &= \mathbf{X}^T \mathbf{W}^{(t)} \underbrace{[\mathbf{X} \beta^{(t)} + \mathbf{u}^{(t)}]}_{\mathbf{z}^{(t)}} \end{aligned}$$

$$\beta^{(t+1)} = (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{z}^{(t)}$$

$$\Rightarrow \text{This is a wls estimate!} \quad \beta^{(t+1)} = \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} (\mathbf{z}^{(t)} - \mathbf{X}\beta)^T \mathbf{W}^{(t)} (\mathbf{z}^{(t)} - \mathbf{X}\beta)$$